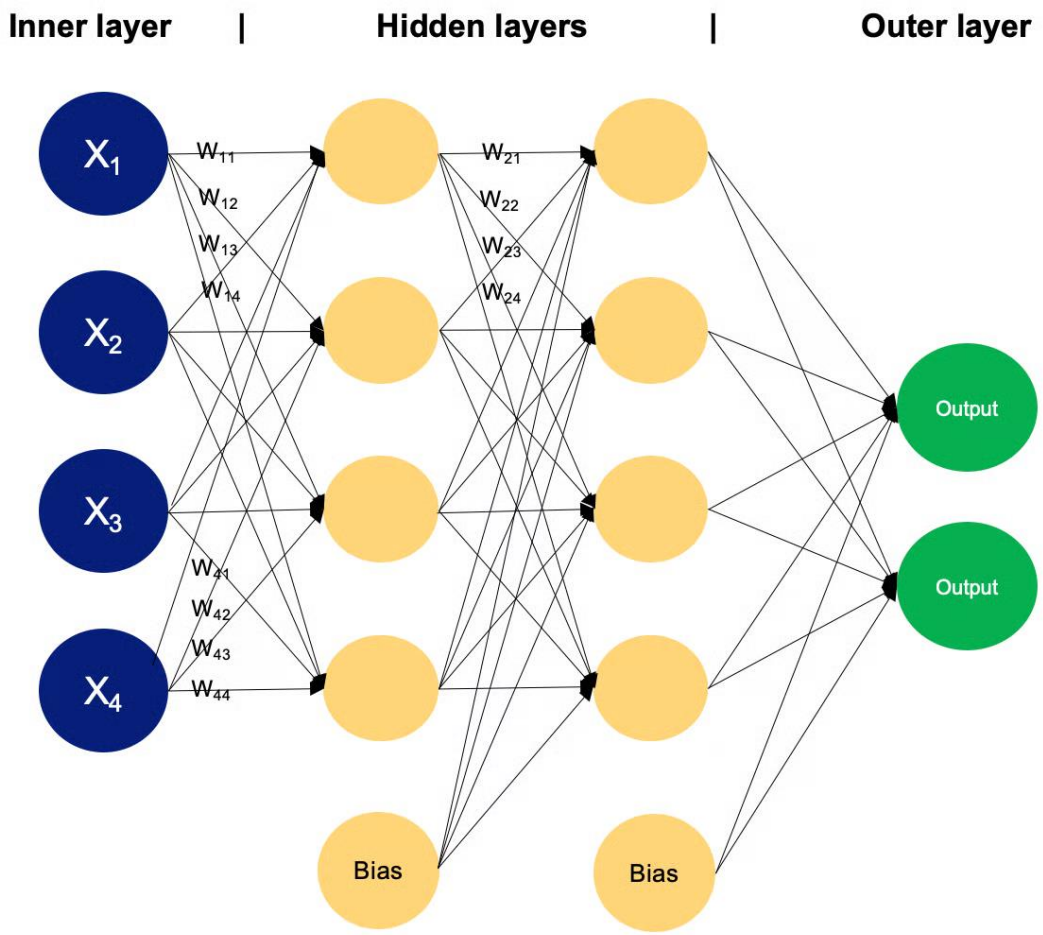
Multi-Layer Perceptron (MLP)

Multi-Layer Perceptron (MLP) consists of fully connected dense layers that transform input data from one dimension to another. It is called multi-layer because it contains an input layer, one or more hidden layers and an output layer. The purpose of an MLP is to model complex relationships between inputs and outputs.

- **Input Layer**: Each neuron or node in this layer corresponds to an input feature. For instance, if you have three input features the input layer will have three neurons.

- **Hidden Layers**: MLP can have any number of hidden layers with each layer containing any number of nodes. These layers process the information received from the input layer.

- **Output Layer**: The output layer generates the final prediction or result. If there are multiple outputs, the output layer will have a corresponding number of neurons.

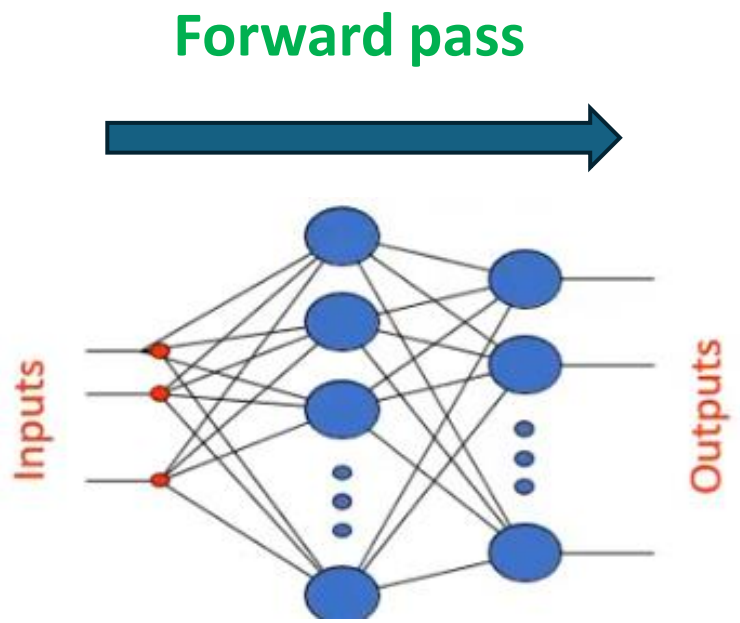


Every connection in the diagram is a representation of the fully connected nature of an MLP. This means that every node in one layer connects to every node in the next layer. As the data moves through the network each layer transforms it until the final output is generated in the output layer.

Working of Multi-Layer Perceptron

- Forward Propagation

In forward propagation the data flows from the input layer to the output layer, passing through any hidden layers.



Each neuron in the hidden layers processes the input as follows:

1. **Weighted Sum**: The neuron computes the weighted sum of the inputs:

$$S = \sum_i w_i x_i + b$$

Where: x_i is the input feature. w_i is the corresponding weight. b is the bias term.

2. **Activation function** : Set Binary, Relu, Sigmoid, Softmax,…

3. **Loss function** is the difference between the predicted value and the actual value for a single data point.

Cost Function: The cost function is the average of the loss functions over the entire training dataset. It provides a single value that summarizes the model's overall performance.

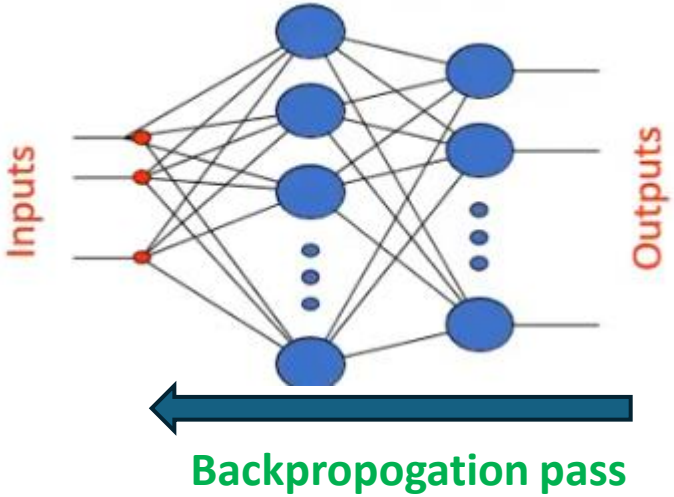
$$L(y, \hat{y}) = \overbrace{\boxed{\frac{1}{n}\sum_{i=1}^{n}}}^{\text{mean}} \overbrace{\boxed{(y_i - \hat{y}_i)}}^{\text{error}} \overbrace{\boxed{^2}}^{\text{squared}}$$

Mean_squared_error

Backpropagation for updating weights

The objective of the training process is to find the weights (W) and biases (b) that minimize the error.

This minimization is achieved through the **gradient descent** algorithm.

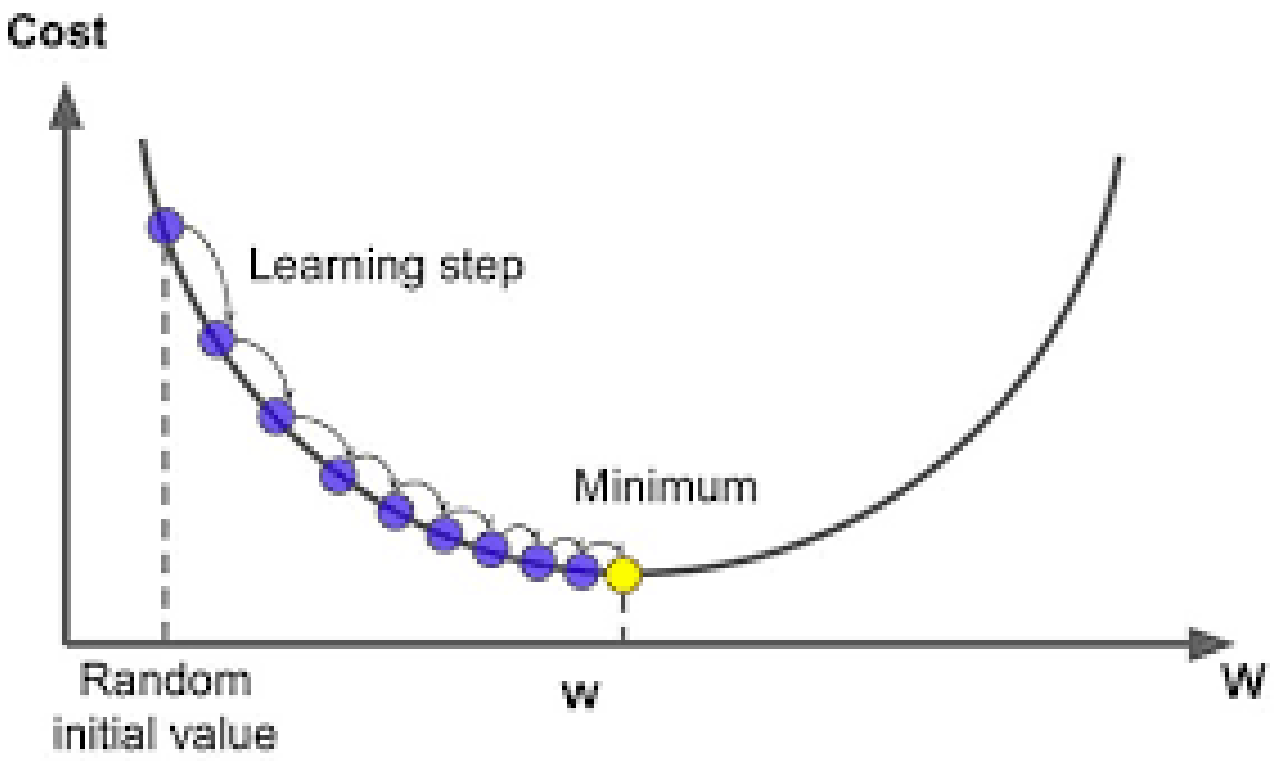


Gradient Descent:

Gradient descent is an optimization algorithm used to minimize the cost function in machine learning models, particularly neural networks. It helps the model find the optimal set of weights that result in the lowest possible error.

Gradient Descent:

To find the best parameters that minimize the error, we use the gradient descent algorithm. The goal is to find the minimum point of the cost function, where the gradient is close to zero (Figure below). The algorithm iteratively moves, step by step, in the direction of steepest descent, toward the minimum point. The step size is named the '**learning rate**', and it is a scalar that determines how much the weights change in each iteration, and therefore how quickly the NN converges. The learning rate is a hyperparameter that has to be tuned. Smaller learning rates require more training time, whereas larger learning rates result in rapid training, but might suffer from compromised performances and instability.

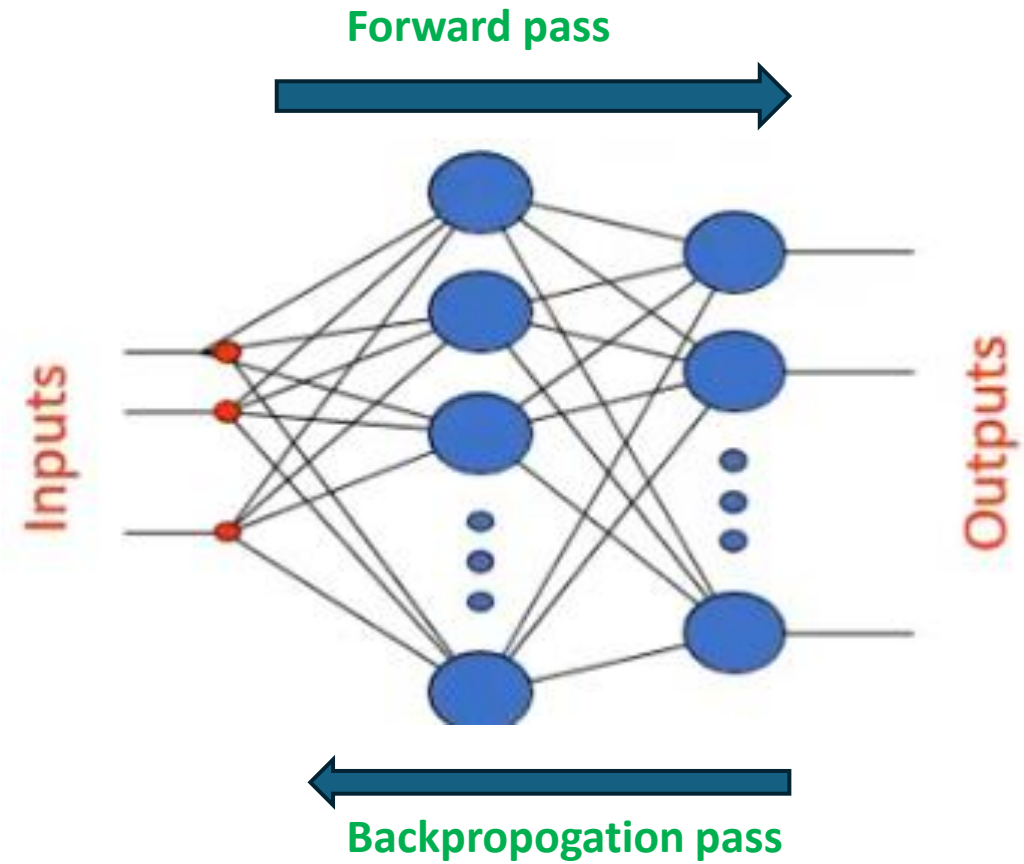

Cost
Learning step
Minimum
Random
initial value
w
W

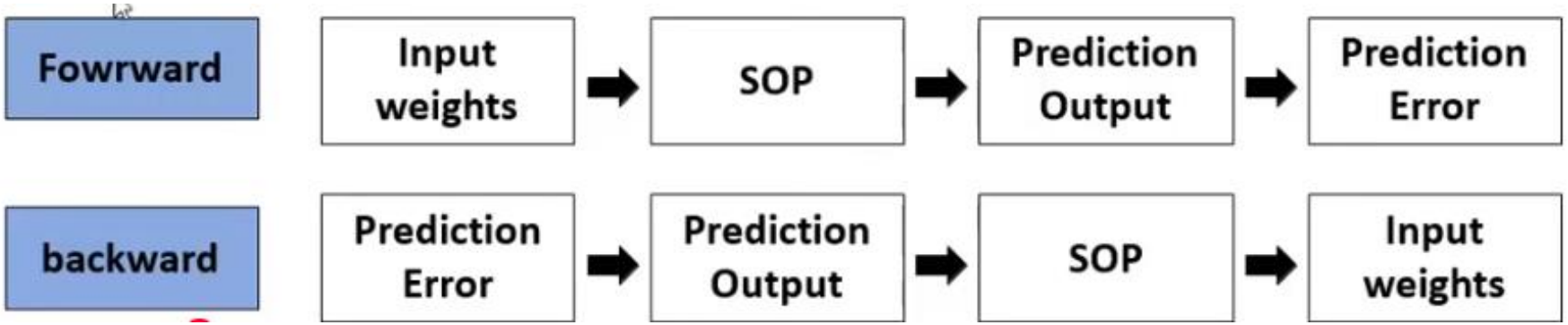
To update the weights, the gradients are multiplied by the learning rate (alpha), and the new weights are calculated by the following formula:

$$W_{new} = W_{old} - \alpha \underbrace{\frac{dJ}{dW}}_{\text{gradient}}$$

W = Weights, alpha = Learning rate, J = Cost.

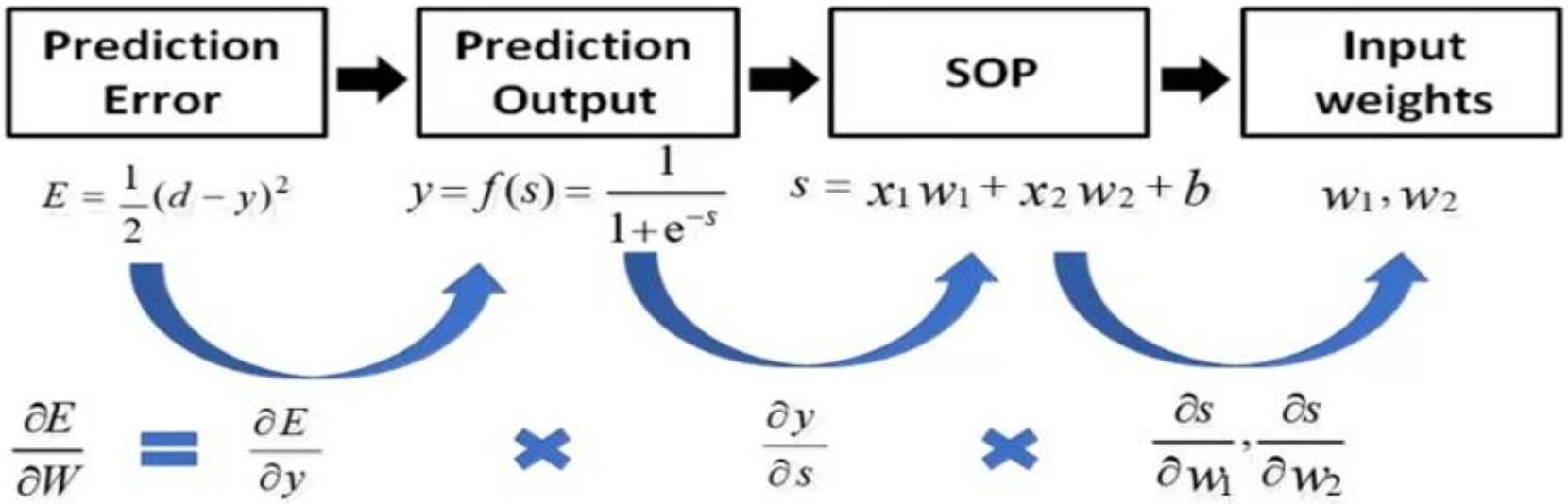
Derivative or gradient : is a slope (En fr**: la pente)** of function at a point. Update of change we want to make in w in code we represent it with ∂w.


Forward pass
Inputs
Outputs
Backpropogation pass


Fowrward
Input weights
SOP
Prediction Output
Prediction Error
backward
Prediction Error
Prediction Output
SOP
Input weights

Chain Rule

Weights derivative



$$\frac{\partial E}{\partial w_1} = \frac{\partial E}{\partial y} x \frac{\partial y}{\partial s} x \frac{\partial s}{\partial w_1}$$

$$\frac{\partial E}{\partial w_2} = \frac{\partial E}{\partial y} x \frac{\partial y}{\partial s} x \frac{\partial s}{\partial w_2}$$

Weights derivative

$$\frac{\partial E}{\partial y} = \frac{\partial(\frac{1}{2}(d-y)^2)}{\partial y} = y - d$$

$$\frac{\partial y}{\partial s} = \frac{\partial(\frac{1}{1+e^{-s}})}{\partial s} = \frac{1}{1+e^{-s}}(1 - \frac{1}{1-e^{-s}})$$

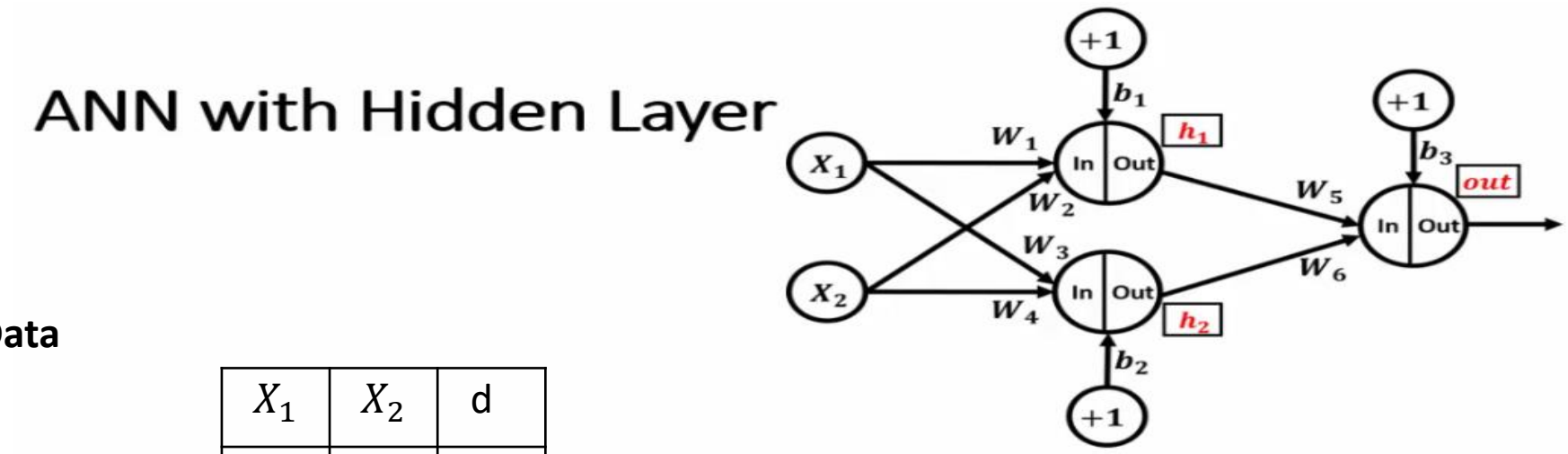
$$\frac{\partial s}{\partial w_1} = \frac{\partial(x_1 w_1 + x_2 w_2 + b)}{\partial w_1} = x_1$$

Or

$$\frac{\partial s}{\partial w_2} = \frac{\partial(x_1 w_1 + x_2 w_2 + b)}{\partial w_2} = x_2$$

$$\boldsymbol{\frac{\partial E}{\partial w_i} = (y-d)\frac{1}{1+e^{-s}}(1 - \frac{1}{1+e^{-s}})x_i}$$

Example



Data

X_1	X_2	d
0.1	0.3	0.03

Initial parameters:

w_1	w_2	w_3	w_4	w_5	w_6	b_1	b_2	b_3
0.5	0.1	0.62	0.2	-0.2	0.3	0.4	-0.1	1.83

Forward Pass – Hidden Layer Neurons

h1 : In

$$h_{1in} = x_1 w_1 + x_2 w_2 + b_1 = 0.1\ \ 0.5\ +\ 0.3\ \ 0.1 + 0.4 = 0.48$$

h1 : Out

$$h_{1out} = \frac{1}{1 + e^{-h_{1in}}} = \frac{1}{1 + e^{-0.48}} = 0.618$$

h2 : In

$$h_{2in} = x_1 w_3 + x_2 w_4 + b_2 = 0.1\ \ 0.62\ +\ 0.3\ \ 0.2 - 0.1 = 0.022$$

h2 : Out

$$h_{2out} = \frac{1}{1 + e^{-h_{2in}}} = \frac{1}{1 + e^{-0.022}} = 0.506$$

Out : In :

$$Out_{in} = h_{1out} w_5 + h_{2out} w_6 + b_3 = 0.618\ (-\ 0.2)\ +\ 0.506\ \ 0.3 + 1.83 = 1.858$$

$$Out_{out} = \frac{1}{1 + e^{-out_{in}}} = \frac{1}{1 + e^{-1.858}} = 0.865$$

Forward pass : Prediction Error

desired =0.03 and predicted $=Out_{out} = 0.865$

$$Error = \frac{1}{2}(desired - out_{out})^2 = \frac{1}{2}(0.03 - 0.865)^2 = 0.349$$

Update the weights :

$$W_{new} = W_{old} - \alpha \underbrace{\frac{dJ}{dW}}_{\text{gradient}}$$

For each weight, we calculate its gradient by : $\frac{\partial E}{\partial w}$

$$So, We\ calculate\ all\ these\ gradients: \frac{dE}{dw_1}, \frac{dE}{dw_2}, \frac{dE}{dw_3}, \frac{dE}{dw_4}, \frac{dE}{dw_5}, \frac{dE}{dw_6}$$

Derivative values: **In the backpropagation pass, you must start from the last layer.** $\frac{dE}{dw_1}, \frac{dE}{dw_2}, \frac{dE}{dw_3}, \frac{dE}{dw_4}, \frac{dE}{dw_5}, \frac{dE}{dw_6}$

$$\frac{dE}{dw_6} = \frac{dE}{dout_{out}} x \frac{dout_{out}}{dout_{in}} \text{x} \frac{dout_{in}}{dw_6}$$

$$\frac{dE}{\partial out_{out}} = \frac{\partial(\frac{1}{2}(desired - out_{out})^2)}{\partial out_{out}} = out_{out} = 0.865 - 0.23 = 0.835$$

$$\frac{dout_{out}}{dout_{in}} = \frac{\partial}{\partial out_{in}}\left(\frac{1}{1+e^{-out_{in}}}\right) = \left(\frac{1}{1+e^{-out_{in}}}\right)\text{x}\left(1 - \frac{1}{1+e^{-out_{in}}}\right) = = \left(\frac{1}{1+e^{-1.858}}\right)\text{x}\left(1 - \frac{1}{1+e^{-1.858}}\right) = 0.23$$

$$\frac{dout_{in}}{dw_6} = \frac{\partial}{\partial w_6}(h_{1out}w_5 + h_{2out}w_6 + b_3) = h_{2out} = 0.506$$

So, $\frac{dE}{dw_6} = \frac{dE}{dout_{out}} x \frac{dout_{out}}{dout_{in}} \text{x} \frac{dout_{in}}{dw_6} = 0.835 \text{x}\ 0.23 \text{x}\ 0.506 = 0.097$

$$\frac{dE}{dw_5} = \frac{dE}{dout_{out}} x \frac{dout_{out}}{dout_{in}} \text{x} \frac{dout_{in}}{dw_5}$$

$$\frac{dE}{\partial out_{out}} = 0.835$$

$$\frac{dout_{out}}{dout_{in}} = 0.23$$

$$\frac{dout_{in}}{dw_5} = \frac{\partial}{\partial w_5}(h_{1out}w_5 + h_{2out}w_6 + b_3) = h_{1out} = 0.618$$

$$\frac{dE}{dw_5} = \frac{dE}{dout_{out}} x \frac{dout_{out}}{dout_{in}} \text{x} \frac{dout_{in}}{dw_5} = 0.835 \text{x} \, 0.23 \text{x} \, 0.618 = 0.119$$

$$\frac{dE}{dw_1} = \frac{dE}{dout_{out}} \text{ x } \frac{dout_{out}}{dout_{in}} \text{ x } \frac{dout_{in}}{dh_{1out}} \text{ x } \frac{dh_{1out}}{dh_{1in}} \text{ x } \frac{dh_{1in}}{dw_1} =$$

$$\frac{dh1_{in}}{dw_1} = \frac{\partial}{\partial w_1}(X_1 w_1 + X_2 w_3 + b_1) = X_1 = 0.1$$

$$\frac{dh_{1out}}{dh_{1in}} \frac{\partial}{\partial h_{1in}} \left(\frac{1}{1+e^{-h_{1in}}}\right) = \left(\frac{1}{1+e^{-h_{1in}}}\right) \text{x} \left(1 - \frac{1}{1+e^{-h_{1in}}}\right) =) = \left(\frac{1}{1+e^{-0.48}}\right) \text{x} \left(1 - \frac{1}{1+e^{-0.48}}\right) = 0.236$$

$$\frac{dout_{in}}{dh_{1out}} = \frac{\partial}{\partial h1_{out}}(h1_{out} w_5 + h2_{out} w_6 + b_3) = w_5 = -0.2$$

$$\frac{dout_{out}}{dout_{in}} = \frac{\partial}{\partial out_{in}} \left(\frac{1}{1+e^{-out_{in}}}\right) = \left(\frac{1}{1+e^{-out_{in}}}\right) \text{x} \left(1 - \frac{1}{1+e^{-out_{in}}}\right) = \left(\frac{1}{1+e^{-1.858}}\right) \text{x} \left(1 - \frac{1}{1+e^{-1.858}}\right) = 0.23$$

$$\frac{dE}{\partial out_{out}} = \frac{\partial(\frac{1}{2}(desired - out_{out})^2)}{\partial out_{out}} = out_{out} = desired - out_{out} = 0.865 - 0.03 = 0.835$$

$$\frac{dE}{dw_1} = 0.835 \text{x } 0.23 \text{x } (-0.2) \text{x} 0.236 \text{x} 0.1 = -0.001$$

$$\frac{dE}{dw_2} = \frac{dE}{dout_{out}} \text{ x } \frac{dout_{out}}{dout_{in}} \text{ x } \frac{dout_{in}}{dh_{1out}} \text{ x } \frac{dh_{1out}}{dh_{1in}} \text{ x } \frac{dh_{1in}}{dw_2} =$$

$$\frac{dh1_{in}}{dw_2} = \frac{\partial}{\partial w2_1}(X_1w_1 + X_2w_2 + b_1) = X_2 = 0.3$$

$$\frac{dh_{1out}}{dh_{1in}} = 0.236$$

$$\frac{dout_{in}}{dh_{1out}} = -0.2$$

$$\frac{dout_{out}}{dout_{in}} = 0.23$$

$$\frac{dE}{\partial out_{out}} = 0.835$$

$$\frac{dE}{dw_2} = 0.835 \text{x } 0.23 \text{x } (-0.2) \text{x} 0.236 \text{x} 0.3 = -0.03$$

$$\frac{dE}{dw_3} = \frac{dE}{dout_{out}} \text{ x } \frac{dout_{out}}{dout_{in}} \text{ x } \frac{dout_{in}}{dh_{2out}} \text{ x } \frac{dh_{2out}}{dh_{2in}} \text{ x } \frac{dh_{2in}}{dw_3} =$$

$$\frac{dh2_{in}}{dw_3} = \frac{\partial}{\partial w_1}(X_1 w_3 + X_2 w_4 + b_1) = X_1 = 0.1$$

$$\frac{dh_{2out}}{dh_{2in}} \frac{\partial}{\partial h_{1in}} \left(\frac{1}{1+e^{-h_{2in}}}\right) = \left(\frac{1}{1+e^{-h_{2in}}}\right) \text{x} \left(1 - \frac{1}{1+e^{-h_{2in}}}\right) =) = \left(\frac{1}{1+e^{-0.022}}\right) \text{x} \left(1 - \frac{1}{1+e^{-0.022}}\right) = 0.25$$

$$\frac{dout_{in}}{dh_{2out}} = \frac{\partial}{\partial h2_{out}}(h1_{out} w_5 + h2_{out} w_6 + b_3) = w_6 = 0.3$$

$$\frac{dout_{out}}{dout_{in}} = 0.23$$

$$\frac{dE}{\partial out_{out}} = 0.865 - 0.03 = 0.835$$

$$\frac{dE}{dw_3} = 0.835 \text{x } 0.23 \text{x } 0.3 \text{x} 0.25 \text{x} 0.1 = 0.001$$

$$\frac{dE}{dw_4} = \frac{dE}{dout_{out}} \text{ x } \frac{dout_{out}}{dout_{in}} \text{ x } \frac{dout_{in}}{dh_{2out}} \text{ x } \frac{dh_{2out}}{dh_{2in}} \text{ x } \frac{dh_{2in}}{dw_4} =$$

$$\frac{dh2_{in}}{dw_4} = \frac{\partial}{\partial w_4}(X_1 w_3 + X_2 w_4 + b_1) = X_2 = 0.3$$

$$\frac{dh_{2out}}{dh_{2in}} \frac{\partial}{\partial h_{1in}} \left(\frac{1}{1+e^{-h_{2in}}}\right) = \left(\frac{1}{1+e^{-h_{2in}}}\right) \text{x} \left(1 - \frac{1}{1+e^{-h_{2in}}}\right) =) = \left(\frac{1}{1+e^{-0.022}}\right) \text{x} \left(1 - \frac{1}{1+e^{-0.022}}\right) = 0.25$$

$$\frac{dout_{in}}{dh_{2out}} = \frac{\partial}{\partial h2_{out}}(h1_{out} w_5 + h2_{out} w_6 + b_3) = w_6 = 0.3 \quad \partial\partial\partial\partial$$

$$\frac{dout_{out}}{dout_{in}} = 0.23$$

$$\frac{dE}{\partial out_{out}} = 0.865 - 0.03 = 0.835$$

$$\frac{dE}{dw_4} = 0.835 \text{x } 0.23 \text{x } 0.3 \text{x} 0.25 \text{x} 0.3 = 0.004$$

$$\frac{dE}{db_3} = \frac{dE}{dout_{out}} x \frac{dout_{out}}{dout_{in}} \text{x} \frac{dout_{in}}{db_3}$$

$$\frac{dE}{\partial out_{out}} = 0.835$$

$$\frac{dout_{out}}{dout_{in}} = 0.23$$

$$\frac{dout_{in}}{db_3} = \frac{\partial}{\partial b_3}(h_{1out}w_5 + h_{2out}w_6 + b_3) = 1$$

$$\frac{dE}{db_3} = \frac{dE}{dout_{out}} x \frac{dout_{out}}{dout_{in}} \text{x} \frac{dout_{in}}{db_3} = 0.835 \text{x} \, 0.23 \text{x} \, 1 = 0.192$$

$$\frac{dE}{db_1} = \frac{dE}{dout_{out}} \text{ x } \frac{dout_{out}}{dout_{in}} \text{ x } \frac{dout_{in}}{dh_{1out}} \text{ x } \frac{dh_{1out}}{dh_{1in}} \text{ x } \frac{dh_{1in}}{db_1} =$$

$$\frac{dh1_{in}}{db_1} = \frac{\partial}{\partial w2_1}(X_1 w_1 + X_2 w_2 + b_1) = 1$$

$$\frac{dh_{1out}}{dh_{1in}} = 0.236$$

$$\frac{dout_{in}}{dh_{1out}} = -0.2$$

$$\frac{dout_{out}}{dout_{in}} = 0.23$$

$$\frac{dE}{\partial out_{out}} = 0.835$$

$$\frac{dE}{db_1} = 0.835 \text{x } 0.23 \text{x } (-0.2) \text{x} 0.236 \text{x} 1 = -0.009$$

$$\frac{dE}{db_2} = \frac{\partial E}{dout_{out}} \text{ x } \frac{dout_{out}}{dout_{in}} \text{ x } \frac{dout_{in}}{dh_{2out}} \text{ x } \frac{dh_{2out}}{dh_{2in}} \text{ x } \frac{dh_{2in}}{db_2} =$$

$$\frac{dh2_{in}}{db_2} = \frac{\partial}{\partial w_4}(X_1 w_3 + X_2 w_4 + b_2) = 1$$

$$\frac{dh_{2out}}{dh_{2in}} = 0.25$$

$$\frac{dout_{in}}{dh_{2out}} = 0.3$$

$$\frac{dout_{out}}{dout_{in}} = 0.23$$

$$\frac{dE}{\partial out_{out}} = 0.835$$

$$\frac{dE}{db_2} = 0.835 \text{x } 0.23 \text{x } 0.3 \text{x} 0.25 \text{x} 1 = 0.014$$

Updated Weights:

w_1	w_2	w_3	w_4	w_5	w_6	b_1	b_2	b_3
0.5	0.1	0.62	0.2	-0.2	0.3	0.4	-0.1	1.83

$$w_1 = w_1 - \propto . \frac{\partial E}{\partial W_1} = 0.5 - 0.01 * (-0.001) = 0.5$$

$$w_2 = w_2 - \propto . \frac{\partial E}{\partial W_2} = 0.1 - 0.01 * (-0.03) = 0.1003$$

$$w_3 = w_3 - \propto . \frac{\partial E}{\partial W_3} = 0.62 - 0.01 * (0.001) = 0.619$$

$$w_4 = w_4 - \propto . \frac{\partial E}{\partial W_4} = 0.2 - 0.01 * (0.004) = 0.199$$

$$w_5 = w_5 - \propto . \frac{\partial E}{\partial W_5} = -0.2 - 0.01 * (0.119) = -0.201$$

$$w_6 = w_6 - \propto . \frac{\partial E}{\partial W_6} = 0.3 - 0.01 * (0.097) = 0.299$$

$$b_1 = b_1 - \propto . \frac{\partial E}{\partial b_1} = 0.4 - 0.01 * (0.009) = 0.399$$

$$b_2 = b_2 - \propto . \frac{\partial E}{\partial b_2} = -0.1 - 0.01 * (0.014) = -0.1$$

$$b_3 = b_3 - \propto . \frac{\partial E}{\partial b_3} = 1.83 - 0.01 * (0.192) = 1.828$$